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Strong NP-hardness of AC power flows feasibility

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ABSTRACT

We prove that testing feasibility for an AC power flow system is a strongly NP-hard problem.
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1. Introduction

The AC-OPF problem is a nonlinear, nonconvex problem arising in the study of electrical networks. This problem has lately received increased attention, spurred by the work in [12]. A recent survey is given in [14]. Also see [8,11,13,15].

An important question concerns the fundamental complexity of AC-OPF. In this setting we note [13], which discusses a proof of weak NP-hardness of AC-OPF on trees; also see [12] for an outline of a proof.

The purpose of this note is to present a proof of strong NP-hardness of AC power flows feasibility testing on general graphs; this proof builds on a section in the PhD thesis [18]. A challenge in the development of such a proof is the fact that the solution to an AC-OPF problem may have irrational coordinates; this necessitates an elaboration in our technique.

We will show that a special case of AC power flows feasibility testing is already strongly NP-hard. The NP-hardness proof is given given in Section 3. Section 2 describes a general version of the AC-OPF problem and develops the NP-hard special case. Some extensions, in particular approximation algorithms in the bounded tree-width case, are discussed in Section 4.

2. The AC-OPF problem

The AC-OPF problem concerns the economic operation of an electrical transmission system (a 'grid') so as to deliver desired amounts of power subject to alternating current (AC) laws. There are many variations of the problem depending on modeling details; we provide a generic but simplified version. The input is a directed graph; we refer to vertices as *buses* and to arcs as *lines*. The formulation is as follows (notation explained below):

$$\text{Min} \sum_i F_i(P_i^g) \tag{1a}$$

Subject to:

$\forall$ line (i,j) :

$$p_{ij:i} = G_{ij}V_i^2 - G_{ij}V_iV_j\cos(\theta_i-\theta_j) - B_{ij}V_iV_j\sin(\theta_i-\theta_j) \tag{1b}$$

$$p_{ij:j} = G_{ij}V_j^2 - G_{ij}V_iV_j\cos(\theta_i-\theta_j) + B_{ij}V_iV_j\sin(\theta_i-\theta_j) \tag{1c}$$

$$q_{ij:i} = -B_{ij}V_i^2 + B_{ij}V_iV_j\cos(\theta_i-\theta_j) - G_{ij}V_iV_j\sin(\theta_i-\theta_j) \tag{1d}$$

$$q_{ij:j} = -B_{ij}V_j^2 + B_{ij}V_iV_j\cos(\theta_i-\theta_j) + G_{ij}V_iV_j\sin(\theta_i-\theta_j) \tag{1e}$$

$$|\theta_i-\theta_j| \le \theta_{ij}^{\max} \tag{1f}$$

$$\max\left\{\sqrt{p_{ij:i}^2+q_{ij:i}^2},\ \sqrt{p_{ij:j}^2+q_{ij:j}^2}\right\} \le U_{ij}^{\max} \tag{1g}$$

$$\max\left\{\sqrt{p_{ij:i}^2+q_{ij:i}^2},\ \sqrt{p_{ij:j}^2+q_{ij:j}^2}\right\} \le U_{ij}^{\max} \tag{1h}$$

$\forall$ bus i :

$$\sum_{(i,j)\in\delta^+(i)} p_{ij:i} + \sum_{(j,i)\in\delta^-(i)} p_{ji:i} = P_i^g - P_i^d \tag{1i}$$

$$\sum_{(i,j)\in\delta^+(i)} q_{ij:i} + \sum_{(j,i)\in\delta^-(i)} q_{ji:i} = Q_i^g - Q_i^d \tag{1j}$$

$$P_i^{g,\min} \le P_i^g \le P_i^{g,\max} \tag{1k}$$

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$$Q_i^{g,\min} \le Q_i^g \le Q_i^{g,\max} \tag{1l}$$
$$V_i^{\min} \le V_i \le V_i^{\max}. \tag{1m}$$

In this formulation, for each line (i, j) we have parameters G_{ij}, B_{ij} describing physical characteristics of the line, and parameters $\theta_{ij}^{\max}$ and $U_{ij}^{\max}$; the operating limits for the line. For every bus i we have parameters $V_i^{\max}, V_i^{\min}$; the minimum and maximum voltage at the bus, parameters $P_i^d, Q_i^d, P_i^{g,\max}, P_i^{g,\min}, Q_i^{g,\max}, Q_i^{g,\min}$; active and reactive power demands and generation limits at the bus, and a function $F_i : \mathbb{R} \to \mathbb{R}$; the generation cost function at bus i.

The optimization problem has the following primary variables, for each bus i: V_i and θ_i (voltage magnitude and phase angle) and P_i^g and Q_i^g (active and reactive power generation at i). Additionally, we have for each line (i, j), auxiliary variables $p_{ij:i}$, $q_{ij:i}$ and $p_{ij:j}$, $q_{ij:j}$ (active and reactive power injected into (i, j) at i and j, respectively).

In the formulation, as is customary, we denote by $\delta^+(i)$ the set of lines of the form (i, j), and likewise with $\delta^-(i)$.

The formulation can be explained as follows. First, constraints (1b)–(1f) describe the underlying *physics*, i.e. how voltage magnitudes and phase angles account for *power flows* on each line; such flows are of two types, "active" and "reactive" (p and q, respectively). Here, $p_{ij:i}$ is the amount of active power injected into line (i, j) at i, or, colloquially, the amount of active power *sent* from i to j along the line (i, j). Likewise, $p_{ij:j}$ represents the (active) power sent from j to i using (i, j). Effectively, we should think of the line (i, j) as an undirected object that is operated in both directions; however the power engineering literature has evolved to use notation that refers to lines as ordered pairs. Note that when $G_{ij} = 0$ we have that $p_{ij:i} = -p_{ij:j}$, or in other words, the amount sent from i to j equals the amount *received* at j. In general, however, $G_{ij} > 0$ (but small) and $p_{ij:i} + p_{ij:j} > 0$ is a "loss". Similar remarks apply to $q_{ij:i}$ and $q_{ij:j}$.

Constraints (1i) and (1j) are flow balance equations. On the right-hand side of (1i), P_i^g (which is a variable) is the active power generated at i, i.e., *injected* into the network at i, whereas P_i^d (a data parameter) is the active power demand at i, is removed from the network. Thus $P_i^g - P_i^d$ is the net injection at i. The reader may be confused by the "+" sign preceding the $\delta^-(i)$ term in the sum: surely that should be a "−"? This apparent error is caused by the convention that for $(j, i) \in \delta^-(i)$, the quantity $p_{ji:i}$ indicates flow departing i rather than entering i. Again, similar considerations apply to (1j).

Finally, the functions F_i in the objective are usually convex and represent generation cost. We can have $P_i^{g,max} = P_i^{g,min} = Q_i^{g,max} = Q_i^{g,min} = 0$, in which case no generation takes place at bus i (and the function F_i is irrelevant). It is possible to have both generation and demand at a common bus, however.

The description we have provided is quite generic, but it skips over a number of modeling details. We refer the reader to the above citations for additional engineering background.

2.1. The special case

We will now construct a special case of AC-OPF in a number of steps. First, we consider the problem of testing feasibility of the system (1b)–(1m). In such a setting the variables P_i^g can be eliminated, and the combination of (1i) and (1k) is replaced by the requirement

$$\gamma_i^{\min} \le \sum_{(i,j)\in\delta^+(i)} p_{ij:i} + \sum_{(j,i)\in\delta^-(i)} p_{ji:i} \le \gamma_i^{\max},$$

where $\gamma_i^{\min} \doteq P_i^{g,\min} - P_i^d$ and $\gamma_i^{\max} \doteq P_i^{g,\max} - P_i^d$. We also assume that for every bus i we have $Q_i^{g,min} = -\infty$ and $Q_i^{g,max} = +\infty$. As a result the Q^g variables and constraints (1j), (1l) can be eliminated.

Next, suppose that for every bus i, $V_i^{\min} = V_i^{\max} = 1$, and that for every line (i, j), $G_{ij} = 0$. Then for bus i, (1b) and (1d) can be rewritten as

$$\forall (i, j) \in \delta^+(i),\ p_{ij:i} = -B_{ij} \sin(\theta_i - \theta_j),$$

and

$$\forall (j, i) \in \delta^-(i),\ p_{ji:i} = B_{ji} \sin(\theta_j - \theta_i).$$

In fact, note that constraints (1b)–(1f) are simply used to define the p and q variables; these constraints can be eliminated and the definitions of the p and q substituted in other constraints.

Let us additionally assume that for every line (i, j), $U_{ij}^{\max} = +\infty$, eliminating constraint (1h). At this point all constraints involving the q variables have been eliminated.

These modifications are common in the literature; they yield a simpler system of equations that preserves some of the important nonlinearities. Denoting, for each line (i, j), $b_{ij} \doteq -B_{ij}$, the simpler system is as follows.

NP-hard feasibility system

∀ bus i:

$$\sum_{(i,j)\in\delta^+(i)} b_{ij} \sin(\theta_i - \theta_j) - \sum_{(j,i)\in\delta^-(i)} b_{ji} \sin(\theta_j - \theta_i) = \gamma_i \tag{2a}$$
$$\gamma_i^{\min} \le \gamma_i \le \gamma_i^{\max}$$

∀ line (i,j):

$$|\theta_i - \theta_j| \le \theta_{ij}^{\max} \tag{2b}$$

Here, the γ_i are auxiliary variables. If we sum all the Eqs. (2a) the left-hand side is zero, and therefore

$$\sum_{\text{buses } i} \gamma_i = 0,$$

or, colloquially, "total supply = total demand".

In the rest of this note we prove that the problem of testing feasibility of a system of type (2) is NP-hard even when the number of bits in the problem description is polynomial in the number of lines and buses. We will assume that for each line (i, j), $0 \le \theta_{ij}^{\max} \le \pi/2$; this quantity will be termed the *maximum phase angle difference* of (i, j). We will also assume that for all (i, j) $b_{ij} > 0$; this parameter is the *susceptance* of (i, j). Both assumptions are common.

We will also use the following notation, for each line (i, j):

$$u_{ij} = \text{limit of } (i, j) \doteq b_{ij} \sin \theta_{ij}^{\max}.$$

Below we will describe a line (i, j) by providing $0 \le u_{ij} \le b_{ij}$ and setting $\theta_{ij}^{\max} = \sin^{-1}(u_{ij}/b_{ij})$ with the proviso that $0 \le \theta_{ij}^{\max} \le \pi/2$.

System (2) is both nonlinear and nonconvex. For background and related problems, see [2–4,6,10,16].

2.1.1. Flow decomposition in the special case

Prior to proceeding with the main proof, we provide a (standard) result that will be useful in the sequel. In system (2) we can interpret, for each line (i, j) the quantity $f_{ij} \doteq b_{ij} \sin(\theta_i - \theta_j)$ as the flow on (i, j), possibly negative. Then (2b) states $|f_{ij}| \le u_{ij}$. Thus where N denotes the node–arc incidence matrix of the underlying network, we will have the familiar flow-balance equations $Nf = \gamma$. Buses i with $\gamma_i > 0$ are *sources* of flow whereas buses i with $\gamma_i < 0$ are *sinks* (to facilitate language, we will occasionally

allow zero-flow sources or sinks). This interpretation is useful because it allows us to rely on the well-known *flow decomposition* theorem [1], as follows.

Definition 1. Consider a system of the form (2) and let (θ, γ) be a feasible solution with at least one nonzero γ_i. A flow decomposition is a family $\{(\mathcal{P}_i, \lambda_i) : 1 \le i \le r\}$ satisfying:

- For $1 \le i \le r$, $\lambda_i > 0$.
- For $1 \le i \le r$, $\mathcal{P}_i$ is a *path* in the network. More precisely, $\mathcal{P}_i$ has the structure $v_{1,i}, v_{2,i}, \ldots, v_{n(i),i}$ where:
  1. Each $v_{j,i}$ is a bus. Further, $v_{1,i}$ is a source and $v_{n(i),i}$ is a sink.
  2. For $1 \le j < n(i)$, either $(v_{j,i}, v_{j+1,i})$ is a line, or $(v_{j+1,i}, v_{j,i})$ is a line. In the first case we say $(v_{j,i}, v_{j+1,i})$ is a *forward* line of $\mathcal{P}_i$ and in the second case we say $(v_{j+1,i}, v_{j,i})$ is a *backward* line of $\mathcal{P}_i$.
- For $1 \le i \le r$, $\theta_{v_{1,i}} > \theta_{v_{2,i}} > \cdots > \theta_{v_{n(i),i}}$.
- For any source k,

  $$\sum(\lambda_i : v_{1,i} = k) = \gamma_k.$$

  For any sink k,

  $$\sum(\lambda_i : v_{n(i),i} = k) = -\gamma_k.$$

- For any line (k, m) we have three alternatives:
  1. If $\theta_k > \theta_m$ then (k, m) is not a backward line of any path $\mathcal{P}_i$ and

  $$b_{km} \sin(\theta_k - \theta_m) = \sum(\lambda_i : (k, m) \text{ a forward line in } \mathcal{P}_i).$$

  2. If $\theta_k < \theta_m$ then (k, m) is not a forward line of any path $\mathcal{P}_i$ and

  $$b_{km} \sin(\theta_m - \theta_k) = \sum(\lambda_i : (k, m) \text{ a forward line in } \mathcal{P}_i).$$

  3. If $\theta_k = \theta_m$ then (k, m) does not belong to any path $\mathcal{P}_i$ (forward or backward).

Lemma 2. *Consider a system of the form (2) and let (θ, γ) be a feasible solution with at least one nonzero γ_i. Then a flow decomposition exists.*

Proof. For any line (i, j) let $f_{ij} = b_{ij} \sin(\theta_i - \theta_j)$. Construct an auxiliary directed graph H and flow vector ϕ on H constructed as follows. For each line (i, j), if $f_{ij} > 0$ then H has the arc (i, j) with $\phi_{ij} = f_{ij}$. And if $f_{ij} < 0$ then H has the arc (j, i) with $\phi_{ji} = -f_{ij}$. Then [1] ϕ has a flow decomposition into directed paths *and* cycles. We will argue that the decomposition will not include any cycles which will conclude the proof. By contradiction, assume $v_0, v_2, \ldots, v_k$ are the vertices of such a cycle, listed in order around the cycle. Then, by construction, for $0 \le j \le k$ we have $\theta_j > \theta_{j+1} \pmod k$, which is impossible. ■

Corollary 3. *Suppose we have solution to a system (2) with a single source, say bus k. Then for any line (k, m) or (m, k) we must have $\theta_k \ge \theta_m$. Likewise, if there is a single sink m, then for any line (k, m) or (m, k) we must have $\theta_k \ge \theta_m$.*

See [4] for related discussions.

3. NP-hardness construction

The NP-hardness proof is structured as follows. In Section 3.1 we will introduce a small example which will be used as a 'gadget' in the main proof, which is given in Section 3.2.

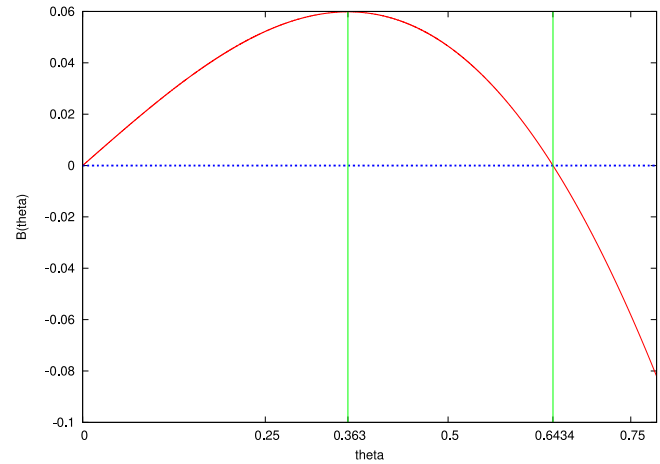

Fig. 1. $\Delta(\theta)$ for $\theta \in [0, \pi/4]$.

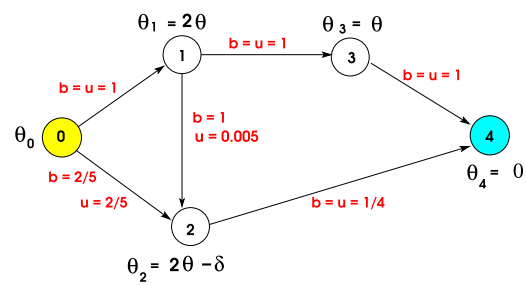

Fig. 2. Basic network B used in proof.

3.1. A special example

Define

$$\Delta(\theta) \doteq -\sin\theta + \frac{5}{8}\sin(2\theta),$$

and set

$$\hat{\theta} = \cos^{-1}\left(\frac{4}{5}\right) \approx .6435,$$

$$\check{\theta} = \cos^{-1}\left(\frac{1}{5} + \sqrt{\frac{1}{25} + \frac{1}{2}}\right) \approx .3630.$$

Then we have:

Lemma 4. *Suppose $0 \le \theta \le \frac{\pi}{2}$. Then:*

(a) *$\Delta(\theta) = 0$ iff $\theta = 0$ or $\theta = \hat{\theta}$.*
(b) *$\Delta'(\theta) > 0$ for $0 \le \theta < \check{\theta}$, $\Delta'(\check{\theta}) = 0$ and $\Delta'(\theta) < 0$ for $\check{\theta} < \theta \le \pi/2$.*

Proof. Part (a) is clear. To prove (b) let $c = \cos\theta$. Then $\Delta'(\theta) = -c + \frac{5}{4}(2c^2 - 1)$ whose only zero in $[0, 1]$ is at $1/5 + \sqrt{1/25 + 1/2}$. ■

See Fig. 1.

Our construction is centered on the network B shown in Fig. 2. In this figure we show, next to each line, its susceptance ("b") and its limit ("u"). The labels next to the buses are values for the phase angles, as explained in points (i)–(iv) below.

We next analyze this network in order to characterize all solutions (θ, γ) to (2a)–(2b) where $\gamma_i = 0$ for all $i \ne 4, 0$. By Lemma 2, either $\theta_0 > \theta_4$ (so 0 is the source and 4 is the sink) or

$\theta_4 > \theta_0$ (so 4 is the source and 0 is the sink), or all flows are zero, i.e. all phase angles are equal.

We focus, first, in the case where $\theta_0 \geq \theta_4$ to show that under these conditions there are two modes of operation: in the first γ_0 is in the range $[0, 0.157)$ and in the second, $\gamma_0 \in [0.77565, 0.886255]$. This well-separated dichotomy will form the basis for our NP-hardness proof.

Let us denote the phase angle at bus i by θ_i, for $0 \leq i \leq 4$. The notation in Fig. 2 is justified as follows.

(i) Without loss of generality $\theta_4 = 0$, because subtracting θ_4 from all angles does not change flows.
(ii) We simplify θ_3 as θ.
(iii) $\theta_1 = 2\theta$ (flow conservation at bus 3).
(iv) We write θ_2 as $2\theta - \delta$. Possibly $\delta < 0$.

We can make some basic observations, assuming 0 is the only source and 4 is the only sink:

1. By Corollary 3, $\theta_3 \geq \theta_4$, and since $\theta_{3,4}^{\max} = \sin^{-1}(1) = \pi/2$, we have $0 \leq \theta \leq \pi/2$.

2. The absolute value of the flow on line $(1, 2)$ is $|\sin\delta|$. But $u_{12} = 0.005$, implying

$$|\sin\delta| \leq 0.005, \quad \text{and} \quad |\delta| < 0.0050001. \tag{3}$$

3. Flow on $(0, 1)$ and $(0, 2)$ must be nonnegative, by Corollary 3, so $\theta_0 \geq \max\{2\theta, 2\theta - \delta\}$. Together with the phase angle limits we have

$$\max\{0, -\delta\} \leq \theta_0 - 2\theta \leq \min\left\{\frac{\pi}{2}, \frac{\pi}{2} - \delta\right\} \tag{4}$$

4. Applying a similar reasoning to line $(2, 4)$ we get

$$\max\{0, \delta\} \;\leq\; 2\theta \;\leq\; \pi/2 + \delta. \tag{5}$$

5. The flow conservation equations at buses 1 and 2 are, respectively,

$$\sin(\theta_0 - 2\theta) = \sin\delta + \sin\theta, \tag{6a}$$

$$\frac{2}{5}\sin(\theta_0 - 2\theta + \delta) = -\sin\delta + \frac{1}{4}\sin(2\theta - \delta) \tag{6b}$$

Let

$$\epsilon_1 \;\doteq\; \sin(\theta_0 - 2\theta + \delta) - \sin(\theta_0 - 2\theta)$$

$$\epsilon_2 \;\doteq\; \sin(2\theta - \delta) - \sin(2\theta).$$

Using these definitions and substituting (6a) into (6b) we obtain

$$\frac{2}{5}\left[\sin\theta + \sin\delta + \epsilon_1\right] = -\sin\delta + \frac{1}{4}\left[\sin(2\theta) + \epsilon_2\right],$$

$$\text{or} \quad \Delta(\theta) = \frac{7}{2}\sin\delta + \epsilon_1 - \frac{5}{8}\epsilon_2.$$

Moreover, $|\epsilon_1| \leq |\delta|$ and $|\epsilon_2| \leq |\delta|$. Hence, from (3)

$$|\Delta(\theta)| \leq 0.02563.$$

6. As a consequence, using Lemma 4 we have that either θ is close to zero or close to θ_0, or more precisely

$$\text{either} \quad 0 \leq \theta \leq 0.1057 \quad \text{or}$$
$$0.578 \leq \theta \leq 0.6952 \; (< \pi/4).$$

We will refer to these two modes of operation as **Mode I** and **II**, respectively.

7. Define **throughput** as the quantity γ_0, i.e. the total flow injected into the system by bus 0, which is the same as $-\gamma_4$, i.e. the total flow received by bus 4. Hence throughput is equal to both quantities

$$\frac{2}{5}\sin(\theta_0 - 2\theta + \delta) + \sin(\theta_0 - 2\theta) \quad \text{and}$$
$$\frac{1}{4}\sin(2\theta - \delta) + \sin(\theta). \tag{7}$$

8. Under Mode I, the second expression in (7) (or (6a)) implies that throughput is at most 0.157. As a result the first expression implies that $\theta_0 < .45$. Also, by (4) $\theta_0 \geq 2\theta$, so under Mode II $\theta_0 > 1$ and using the second expression in (7), throughput is in the range $[0.77565, 0.886255]$.

The above observations imply the following:

Summary 3.1. Network B operation

When $\theta_0 \geq \theta_4$ and $\gamma_i = 0$ for all $i \neq 0, 4$, there are two modes of operation for network B. Defining

$$\hat{\theta} = \cos^{-1}\left(\frac{4}{5}\right) \approx 0.644,$$

$$T^* \;\doteq\; \frac{1}{4}\sin(2\hat{\theta}) + \sin(\hat{\theta}) \;=\; 0.84, \tag{8}$$

(1) In **Mode I**, $0 \leq \theta_0 < \frac{1}{2}$, and throughput is in the range $\left[0, \frac{T^*}{5}\right]$. The solution given by $\theta_0 = \theta = \delta = 0$ (all angles at zero) has zero throughput and is in this mode.
(2) In **Mode II**, $1 < \theta_0$, and throughput is at least $\frac{9}{10}T^*$. The solution

$$\theta_0 = 3\hat{\theta}, \;\; \theta_1 = \theta_2 = 2\hat{\theta}, \;\; \theta_3 = \hat{\theta}, \;\; \theta_4 = 0 \tag{9}$$

(so $\delta = 0$) is easily verified as feasible. Its throughput is precisely T^*.

When $\theta_0 < \theta_4$ and $\gamma_i = 0$ for all $i \neq 0, 4$, bus 4 is the only source and 0 is the only sink.

3.2. NP-hardness construction

Here we consider the following problem:

ACFEAS: Given a system of constraints (2), is there a feasible solution?

We will show that this problem is strongly NP-hard using a reduction from ONE-IN-THREE 3-SAT, defined as follows:

ONE-IN-THREE 3-SAT: Given clauses $C_1, \ldots, C_m$ on boolean variables $x_1, \ldots, x_n$, where each C_i uses three literals, is there a truth assignment to the x_j where in each clause there is precisely one true literal?

The construction will rely on several adaptations of the system B considered in the previous section. In what follows, we let S, H be rational values such that (see (8))

$$T^* - \frac{1}{(m+n)^2} \;\leq\; S \;\leq\; T^* \leq H$$
$$\leq\; T^* + \frac{1}{(m+n)^2}. \tag{10}$$

Such rationals can be written using a number of bits polynomial in $m + n$ [7].

We assume we are given an instance, **I**, of ONE-IN-THREE 3SAT with notation as above. For each variable x_j, we first construct a "variable" network $V(j)$ using two copies of the B network, denoted B_{x_j} and $\mathrm{B}_{\bar{x}_j}$, respectively plus three additional buses, s_{x_j}, $s_{\bar{x}_j}$ and t_j. See Fig. 3. Let $0_{x_j}, 1_{x_j}, \ldots, 4_{x_j}$ (resp., $0_{\bar{x}_j}$ through $4_{\bar{x}_j}$) denote the copies of buses 0 through 4 in B_{x_j} ($\mathrm{B}_{\bar{x}_j}$). We have lines

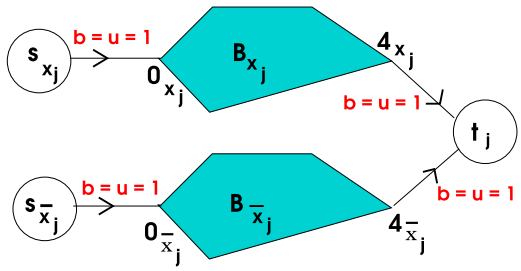

Fig. 3. Network $V(j)$.

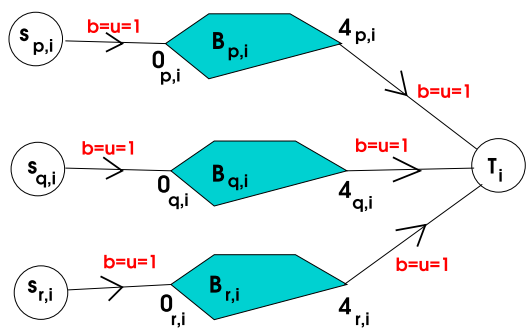

Fig. 4. Network $C(i)$ from clause $C_i = (p \vee q \vee r)$.

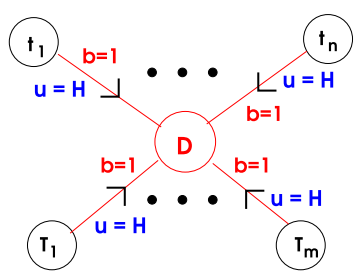

Fig. 5. Attaching bus D.

$(s_{x_j}, 0_{x_j})$, $(s_{\bar{x}_j}, 0_{\bar{x}_j})$, $(4_{x_j}, t_j)$, $(4_{\bar{x}_j}, t_j)$, all with susceptance and limit equal to 1.

Next, for each clause C_i we construct a "clause" network $C(i)$. Suppose $C_i = (p \vee q \vee r)$. The network $C(i)$ will contain three copies of the B network, denoted $\mathrm{B}_{p,i}$, $\mathrm{B}_{q,i}$ and $\mathrm{B}_{r,i}$. See Fig. 4.

The copy of each bus k of B, in $\mathrm{B}_{p,i}$, is labeled $k_{p,i}$ and likewise with $\mathrm{B}_{q,i}$ and $\mathrm{B}_{r,i}$. There are additional buses $s_{p,i}$, $s_{q,i}$ and $s_{r,i}$, connected to $0_{p,i}$, $0_{q,i}$, and $0_{r,i}$, respectively, and a bus T_i connected from $4_{p,i}$, $4_{q,i}$, and $4_{r,i}$. These new lines have unit susceptance and limit.

The variable networks and clause networks are assembled together as follows.

(a) There is an additional bus, D. For $1 \le j \le n$ we have the line (t_j, D). For $1 \le i \le m$ we have the line (T_i, D). All of these lines have susceptance 1 and limit H. See Fig. 5.

(b) For each variable x_j $(1 \le j \le n)$ we have four additional buses, denoted L_{x_j}, R_{x_j}, $L_{\bar{x}_j}$ and $R_{\bar{x}_j}$ respectively. Bus L_{x_j} (resp., R_{x_j}) is connected to 0_{x_j} (4_{x_j}), and, for any clause C_i containing x_{x_j}, bus L_{x_j} (resp., R_{x_j}) is connected $0_{x_j,i}$ $(4_{x_j,i})$. Likewise, bus $L_{\bar{x}_j}$ (resp., $R_{\bar{x}_j}$) is connected to $0_{\bar{x}_j}$ $(4_{\bar{x}_j})$, and, for any clause C_i containing $\bar{x}_j$, bus $L_{\bar{x}_j}$ (resp., $R_{\bar{x}_j}$) is connected $0_{\bar{x}_j,i}$ $(4_{\bar{x}_j,i})$. All lines mentioned here have limit $T^*/1000$ and unit susceptance. See Fig. 6 for an example.

To complete the network description we define the quantities $\gamma^{\min}$ and $\gamma^{\max}$ as follows. First, set

$$\gamma_D^{\min} = -(m+n)H \text{ and } \gamma_D^{\max} = -(m+n)S$$

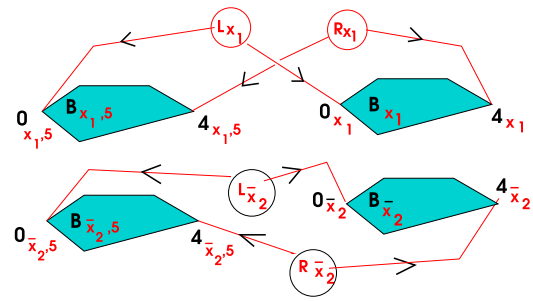

Fig. 6. Partial assembly in the case $C_5 = (x_1 \vee \bar{x}_2 \vee x_3)$; ($B_{x_3}$ not shown).

(recall Eq. (10)). Next, for $1 \le j \le n$,

$$\gamma_{s_{x_j}}^{\min} = \gamma_{s_{\bar{x}_j}}^{\min} = 0 \quad \text{and} \quad \gamma_{s_{x_j}}^{\max} = \gamma_{s_{\bar{x}_j}}^{\max} = H.$$

Next, for $1 \le i \le m$, writing $C_i = (p \vee q \vee r)$),

$$\gamma_{s_{p,i}}^{\min} = \gamma_{s_{q,i}}^{\min} = \gamma_{s_{r,i}}^{\min} = 0 \quad \text{and}$$
$$\gamma_{s_{p,i}}^{\max} = \gamma_{s_{q,i}}^{\max} = \gamma_{s_{r,i}}^{\max} = H,$$

Finally, for all remaining buses k, $\gamma_k^{\min} = \gamma_i^{\max} = 0$.

This concludes the description of the system (2). For brevity, we will refer to this instance of ACFEAS as **AC(I)**. Note that in any feasible solution bus D will be the only sink, whereas the s_j and $s_{x_j,i}$ buses are the only possible sources. Clearly, the number of bits needed to describe **AC(I)** is polynomial in $n + m$.

Lemmas 10 and 11 given below establish the NP-hardness result. In Lemma 10 we show that if there is a feasible solution to **AC(I)**, then in each network $V(j)$ precisely one of the two copies of network B operates in Mode II—this feature is used as a "truth assignment". Moreover the truth assignments satisfy the clauses, i.e. if, say we have a clause $C_i = (x_h \vee x_j \vee x_k)$, and B_h operates in Mode II, then $\mathrm{B}_{x_h,i}$ operates in Mode II, also. Further, in each network $C(i)$ precisely one of the three copies of network B operates in mode II. Prior to presenting the proof, we provide a short paragraph with intuition as to how this strategy is attained.

First we will argue that every line incident with bus D will carry flow $\approx T^*$ (because the demand of D is approximately $(n+m)T^*$ and the $(n+m)$ lines all have limit approximately T^*). Next, consider an index $1 \le j \le n$. Then flow conservation at bus t_j implies that at least one of the two arcs $(4_{x_j}, t_j)$, $(4_{\bar{x}_j}, t_j)$ must carry flow at least $T^*/2$. Say it is the first. Then we observe that the arc $(4_{x_j}, R_{x_j})$ has very small limit. This implies that bus 4_{x_j} must operate as a sink within network B_{x_j}, with demand at least $T^*/2$ minus a very small correction. But then B_{x_j} operates in Mode II, and in fact the flow reaching 4_{x_j} is at least $.9T^*$ – but in that case the flow on $4_{\bar{x}_j}$ is *small* – it is at most $T^* - .9T^*$, which means that network $\mathrm{B}_{\bar{x}_j}$ *cannot* be operating in Mode II. Conclusion: precisely one of the two networks B_{x_j}, $\mathrm{B}_{\bar{x}_j}$ operates in Mode II. Likewise, one proves that for any clause $C_i = (p \vee q \vee r)$, precisely one of the three networks $\mathrm{B}_{p,i}$, $B_{q,i}$, $B_{r,i}$ operates in Mode II. Finally, because all lines incident with buses L_p, R_p have very small limit, then the phase angles at the endpoints of each such line are nearly equal. Thus, if some literal p such that B_p operates in Mode II appears in some clause C_i, it will be the case that $\mathrm{B}_{p,i}$ must also operate in Mode II, and thus it is the one copy of B within $C(i)$ that operates in Mode II. The detailed proof is given next.

We begin with some observations. In Lemmas 5–9 we assume that there is feasible solution (θ, γ) to **AC(I)**, and we denote by $f_{uv} = b_{uv} \sin(\theta_u - \theta_v)$ the flow on any line (u, v).

Lemma 5. *Let $1 \le j \le n$. Let p be either x_j or $\bar{x}_j$. Then (a) $f_{4_p,t_j} \ge -T^*/500$. (b) Suppose $f_{4_p,t_j} \ge T^*/4$. Then network B_p operates in Mode II. (c) Suppose network B_p operates in Mode II. Then $f_{4_p,t_j} \ge .9T^* - T^*/1000$.*

Proof. (a) By adding the balance equations (2a) over all buses of B_p we obtain

$$f_{4_p,t_j} = f_{s_p,0_p} + f_{L_p,0_p} + f_{R_p,4_p}.$$

The result follows because $f_{s_p,0_p} \ge 0$ (since $\gamma_{s_p}^{\min} = 0$) and the limit for lines $(L_p, 0_p)$ and $(R_p, 4_p)$ is $T^*/1000$. (b) Suppose $f_{4_p,t_j} \ge T^*/4$. Since $\gamma_{4_p}^{\min} = \gamma_{4_p}^{\max} = 0$, the flow balance equation at bus 4_p is

$$f_{R_p,4_p} + f_{3_p,4_p} + f_{2_p,4_p} = f_{4_p,t_j},$$

and given that $u_{R_p,4_p} = T^*/1000$, we have $f_{3_p,4_p} + f_{2_p,4_p} \ge T^*/4 - T^*/1000$. In other words, if we consider the disposition of flows within B_p, bus 4_p is a *sink*, with demand at least $T^*/4 - T^*/1000$. Hence B_p operates in Mode II, as desired (refer to Summary Summary 3.1). (c) In Mode II, the sink receives flow at least $.9T^*$. The result follows as in (b). ■

Lemma 6. *Let $1 \le i \le m$, and suppose $C_i = (p \vee q \vee r)$. Then: (a) $f_{4_{p,i},T_i}$, $f_{4_{q,i},T_i}$ and $f_{4_{r,i},T_i}$ are all at least $-T^*/500$. (b) For $h = p, q,$ or r, suppose $f_{4_{h,i},T_i} \ge T^*/4$. Then network $B_{h,i}$ operates in Mode II. (c) For $h = p, q,$ or r suppose network $B_{h,i}$ operates in Mode II. Then $f_{4_{h,i},T_i} \ge .9T^* - T^*/1000$.*

Proof. Similar to that of Lemma 5 and will be omitted. ■

Lemma 7. *The flow on every line that ends at D has value at least $T^* - \frac{1}{(m+n)^2}$.*

Proof. Let ϕ be the smallest flow attained by any line that ends at D. Since there are $n+m$ such lines in total, and they all have limit at most $T^* + \frac{1}{(m+n)^2}$, and $\gamma_D^{\min} \ge (n+m)(T^* - \frac{1}{(m+n)^2})$, we have

$$(n+m-1)\left(T^* + \frac{1}{(m+n)^2}\right) + \phi \ge (n+m)\left(T^* - \frac{1}{(m+n)^2}\right),$$

from which the result follows. ■

Lemma 8. *(a) Let $1 \le i \le m$, and suppose $C_i = (p \vee q \vee r)$. Then precisely one of $B_{p,i}$, $B_{q,i}$ and $B_{r,i}$ operates in Mode II. (b) Let $1 \le j \le n$. Then precisely one of B_{x_j}, $B_{\bar{x}_j}$ operates in Mode II.*

Proof. (a) Since $\gamma_{T_i}^{\min} = \gamma_{T_i}^{\max} = 0$, the flow balance equation at bus T_i is given by

$$f_{4_{p,i},T_i} + f_{4_{q,i},T_i} + f_{4_{r,i},T_i} = f_{T_i,D} \tag{11}$$

By Lemma 7, $f_{T_i,D} \ge T^* - \frac{1}{(m+n)}$. But then (11) implies that

$$\max\{f_{4_{p,i},T_i}, f_{4_{q,i},T_i}, f_{4_{r,i},T_i}\} \ge \frac{1}{3}\left(T^* - \frac{1}{m+n}\right) > T^*/4.$$

Without loss of generality, the maximum is attained by $f_{4_{p,i},T_i}$. Then Lemma 6(b) implies that $B_{p,i}$ operates in Mode II. Lemma 6 (c) then implies that $f_{4_{p,i},T_i} > .9T^* - T^*/1000$. Hence from (11) we obtain

$$f_{4_{q,i},T_i} + f_{4_{r,i},T_i} \le f_{T_i,D} - .9T^* + T^*/1000 \le H - .9T^* + T^*/1000 \le T^*/9 + \frac{1}{(m+n)^2}.$$

By Lemma 6(a), the maximum of $f_{4_{q,i},T_i}$ and $f_{4_{r,i},T_i}$ is less than $T^*/4$. It follows from Lemma 6(d) that neither $B_{q,i}$ not $B_{r,i}$ operate in Mode II. (b) Proof similar to (a). ■

Lemma 9. *Let $1 \le i \le m$, and suppose $C_i = (x_p \vee x_q \vee x_r)$ (a) Suppose B_p operates in Mode II. Then $B_{p,i}$ operates in Mode II also, and neither $B_{q,i}$ nor $B_{r,i}$ operate in Mode II. (b) If B_p does not operate in Mode II then neither does $B_{p,i}$.*

Proof. (a) Since B_p operates in Mode II, $\theta_{0_p} - \theta_{4_p} > 1$. The lines incident with L_p and R_p have unit susceptance and limit $T^*/1000$. It follows that $\theta_{0_{p,i}} - \theta_{4_{p,i}} > 1 - 4T^*/1000$. Hence $B_{p,i}$ also operates in Mode II as well. Applying Lemma 8(a) we conclude that neither $B_{q,i}$ nor $B_{r,i}$ operate in Mode II. (b) If B_{x_j} does not operate in Mode II then $\theta_{0_j} - \theta_{4_j} < .5$. A similar argument as in (a) concludes the proof. ■

Now we can conclude as desired:

Lemma 10. *Suppose there is a feasible solution to instance* **AC(I)**. *Then the instance* **I** *of ONE-IN-THREE 3SAT is satisfiable.*

Proof. By Lemma 8(b), for $1 \le j \le n$ one of the two networks B_{x_j}, $B_{\bar{x}_j}$ operates in Mode II. In the first case we set $x_j =$ true and in the other $\bar{x}_j =$ true. The result now follows by Lemmas 8(a) and 9. ■

Next we prove the converse to Lemma 10, that is to say, we will show that if instance **I** of ONE-IN-THREE 3SAT is satisfiable then instance **AC(I)** of ACFEAS is feasible. This will suffice to complete the NP-hardness proof.

Important: The solution to **AC(I)** that we construct will take irrational values—yet it is the *existence* of such a solution that matters towards our proof.

Let us assume that there is a truth assignment $\mathcal{A}$ that satisfies instance **I** of ONE-IN-THREE 3SAT. In short, the solution to **AC(I)** that we construct will be such that for each true literal p (i.e. x_j or $\bar{x}_j$ for some j), network B_p operates in Mode II, and there are T^* units of flow routed from s_p through, B_p, to t_j and then to D. Similarly for each clause C_i that includes p as a term, there will be T^* units of flow routed from $s_{p,i}$, through $B_{p,i}$ (operated in Mode II), to T_i and then to D. Phase angles at remaining buses will be set up so that all other flows will be zero.

Now we describe the construction in detail. Write

$$\theta^* \doteq \sin^{-1}(T^*).$$

We construct phase angles θ as follows. First, set

$$\theta_{t_j} = -\theta^* \ (1 \le j \le n), \tag{12a}$$

$$\theta_{T_i} = -\theta^* \ (1 \le i \le m), \tag{12b}$$

$$\theta_D = -2\theta^*. \tag{12c}$$

Note that this assignment implies that there are T^* units of flow on each arc in $\delta^-(D)$, for a total of $(n+m)T^*$ units, as desired.

Next consider any literal p that is **true** under $\mathcal{A}$. Then we set (recall Eq. (9)):

$$\theta_{s_p} = \theta^* + 3\hat{\theta}, \quad \theta_{L_p} = \theta_{0_p} = 3\hat{\theta}, \tag{13a}$$

$$\theta_{1_p} = \theta_{2_p} = 2\hat{\theta}, \quad \theta_{3_p} = \hat{\theta}, \tag{13b}$$

$$\theta_{4_p} = 0 = \theta_{R_p}. \tag{13c}$$

Moreover, for each clause C_i that includes p as a term then we also operate $B_{p,i}$ we set:

$$\theta_{s_{p,i}} = \theta^* + 3\hat{\theta}, \ \theta_{0_{p,i}} = 3\hat{\theta}, \tag{14a}$$

$$\theta_{1_{p,i}} = \theta_{2_{p,i}} = 2\hat{\theta},\ \theta_{3_{p,i}} = \hat{\theta},\ \theta_{4_{p,i}} = 0. \tag{14b}$$

Assignments (13) imply that B_p operates in Mode II, with flow of T^* units on $(s_p, 0_p)$, also T^* units of flow routed from 0_p to 4_p within B_p, and then (using (12)) T^* units on $(4_p, t_j)$, where j is the index such that $p = x_j$ or $\bar{x}_j$. Moreover, the flow on $(L_p, 0_p)$ and $(R_p, 4_p)$ equals zero. Likewise, (14) implies we will have T^* units of flow on $(s_{p,i}, 0_{p,i})$, also T^* units of flow routed from $0_{p,i}$ to $4_{p,i}$ within $B_{p,i}$ and finally (using (12)), T^* units on $(4_{p,i}, T_i)$. Moreover, the flow on $(L_p, 0_{p,i})$ and $(R_p, 4_{p,i})$ equals zero.

Consider now the case of a literal that is **false** under $\mathcal{A}$. Say literal p is false. Then all buses in B_p, as well as s_p, L_p and R_p are assigned phase angles $-\theta^*$. Thus all flows within B_p are zero, as well as the flows on $(s_p, 0_p)$, $(L_p, 0_p)$, $(R_p, 4_p)$ and (using (12)) on $(4_p, t_k)$, where $p = x_k$ or $\bar{x}_k$.

Moreover, for each clause C_i that includes p as a term, all buses in $B_{p,i}$, as well as $s_{p,i}$, are given phase angle $-\theta^*$. Thus the flow on $(s_{p,i}, 0_{p,i})$ equals zero, as is the case for all lines within $B_{p,i}$ and for lines $(L_p, 0_{p,i})$ and $(R_p, 4_{p,i})$. Finally the flow on $(4_{p,i}, T_i)$ also equals zero, by (12).

This concludes the description of the phase angle assignments. Note that for any j, $1 \le j \le n$ one of x_j and $\bar{x}_j$ is true. So one of the two arcs $(4_{x_j}, t_j)$ and $(4_{\bar{x}_j}, t_j)$ carries flow T^* and the other carries zero flow.

Finally, for each clause $C_i = (p \vee q \vee r)$ exactly one of p, q, r is true: thus precisely one of the arcs $(4_{p,i}, T_i)$, $(4_{q,i}, T_i)$, $(4_{r,i}, T_i)$ carries flow T^* while the other two carry zero flow. Hence in total T^* units of flow enter T_i.

In conclusion, we have proved:

Lemma 11. *Suppose* **I** *is satisfiable. Then there is a feasible solution to* **AC(I)**. ■

4. Approximation algorithms

As discussed above [13] shows that testing AC feasibility is hard already on *trees*. This leaves open the possibility of pseudo-polynomial time algorithms for approximate feasibility checking (and approximate optimization) on appropriately sparse networks. Here we will outline a positive answer for the general formulation (1), on graphs with fixed *tree-width* [17]. Tree-width is a parameterization of structurally sparse graphs; forests are the graphs with tree-width 1. For more details see [5].

To obtain these results we use the well-known reformulation of AC power flow problems that replaces each pair V_i, θ_i of variables with the real and imaginary components of the complex number $V_i e^{\sqrt{-1}\theta_i}$. Let those two components be e_i and f_i, respectively. In that case the expression for $p_{ij,i}$, (1b), becomes

$$\begin{aligned} p_{ij,i} \ =\ & G_{ij}(e_i^2 + f_i^2) - G_{ij}(e_i e_j + f_i f_j) \\ & + B_{ij}(e_i f_j - f_j e_i). \end{aligned} \tag{15}$$

Similar expressions can be obtained for $q_{ij,i}$, $p_{ij,j}$ and $q_{ij,j}$.

Using this representation, constraint (1i) (and also (1j)) have a special structure. Each such constraint becomes a sum of quadratic expressions. There is a term of the sum (i.e., a quadratic expression) for each line in $\delta^+(i)$ and a term for each line in $\delta^-(i)$. The term arising from a given line (i, j), like (15), involves the variables e_i, f_i and e_j, f_j and likewise for each term arising from a line (j, i).

Clearly, the same is true for constraint (1h) (written as two separate constraints and with the square root, squared). It is also true for (1m) (write it as $(V_i^{\min})^2 \le e_i^2 + f_i^2 \le (V_i^{\max})^2$; the sum is vacuous). It is even true for (1g), since we assume that $\theta_{ij}^{\max} \le \pi/2$, using the arctangent function. This step introduces an irrational quantity which has to be approximated [9].

In summary, thus, system (1) is a special case of the following setup: we are given a graph G, possibly with parallel edges. For each vertex u there is a set of variables $\mathcal{X}_u$ associated with k. Moreover for each edge $\{u, v\}$ of the graph we are given sets P_{uv} of polynomials in variables from $\mathcal{X}_u \cup \mathcal{X}_v$. Finally, we are given a set of polynomial constraints: each constraint is associated with some vertex u, and it is a sum of terms, one per each edge $\{u, v\}$ in $\delta(u)$. Each such term is a polynomial in the family P_{uv}.

A linear-objective optimization problem with this type of constraint setup is termed an NPO (network polynomial optimization problem). In [5] the following result is proved:

Theorem 1.6 ([5]). *Consider an NPO over a graph G with maximum degree D, and with at most Δ variables and constraints associated with any vertex. Let ρ be the maximum degree of any polynomial in the families P_{uv}. Finally, let $0 < \epsilon < 1$. Then there is a* linear program *with the following attributes:*

1. *The number of variables and constraints is $O((D\rho/\epsilon)^{O(\Delta\omega)}\, n \log(\rho/\epsilon))$; here ω is the tree-width of G.*
2. *Any feasible solution to the LP violates any constraint of the NPO by $O(C\epsilon)$, where C is the largest 1-norm of the coefficients of any constraint.*
3. *The optimality error of the LP is at most $O(K\epsilon)$, where K is the 1-norm of the objective coefficients.*

One can further prove that the dependence on C (in the feasibility error) is best possible—removing it cannot be done unless $P = NP$.

To understand the relevance of this problem toward this paper, consider an AC feasibility checking problem over a graph with tree-width ω and maximum degree D. Then $\Delta \le 5$ and $\rho = 2$. We conclude that Theorem 1.6 yields an LP of size polynomial in the network parameters and polynomial in $(D/\epsilon)^{\omega}$, i.e., for fixed ω it is polynomial in the network parameters and pseudopolynomial in $1/\epsilon$. The LP yields solutions that are *approximately* feasible: the feasibility error is $O(C\epsilon)$, where $C = O(\Delta)$ times the largest parameter $G_{ij}, G_{ji}, B_{ij}, B_{ji}$.

It would be of interest to develop practical versions of the LP formulation just described; verifying infeasibility of AC power flow formulations can be a challenging problem.

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